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1. Introduction

Network scanning is a fundamental technique used in cybersecurity to identify active hosts, open ports, and running services on a network. It helps security professionals understand the attack surface of a system and identify potential vulnerabilities.

Advanced scanning techniques provide deeper insight into how systems respond to network probes and allow analysts to detect misconfigurations, exposed services, and weak security settings.

In this project, advanced scanning techniques were performed using Nmap on a vulnerable testing machine in a controlled lab environment.

2. Objective

The objective of this project is:

- To perform advanced network scanning techniques
- To identify open ports and running services
- To analyze potential security risks associated with exposed services
- To understand how attackers gather information about target systems

3. Tools Used

Tool	Description
Kali Linux	Penetration testing operating system
Nmap	Network scanning and security auditing tool
Metasploitable2	Vulnerable virtual machine used for testing

Lab Setup

The experiment was conducted in a controlled virtual lab environment.

Machine	IP Address
Kali Linux	192.168.150.178
Metasploitable2	192.168.150.179

Kali Linux was used as the attacker machine, while Metasploitable2 acted as the target system.

5. Methodology

The following steps were followed:

1. Configure the lab environment.
2. Verify connectivity between Kali and Metasploitable.
3. Perform basic network scanning.
4. Execute advanced Nmap scanning techniques.
5. Analyze discovered open ports and services.

6. Commands / Techniques Used

Figure 1 (4 screenshots)

Network setup

- Kali IP
- Metasploitable IP
- Ping Kali
- Ping target

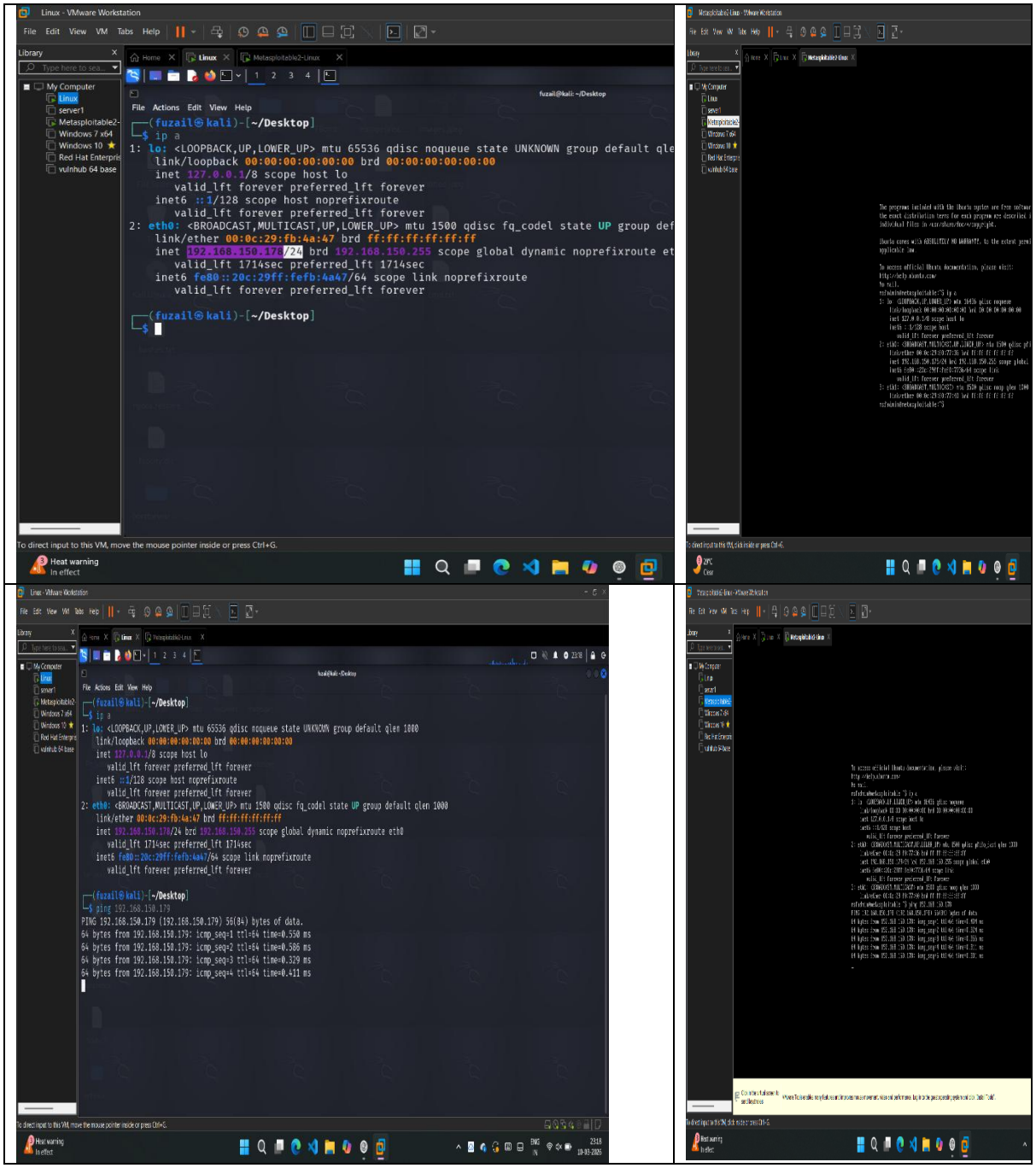


Figure 2: Command: nmap 192.168.150.179

Purpose: Identify open ports on the target system. Basic Nmap scan identifying open ports on the target machine.

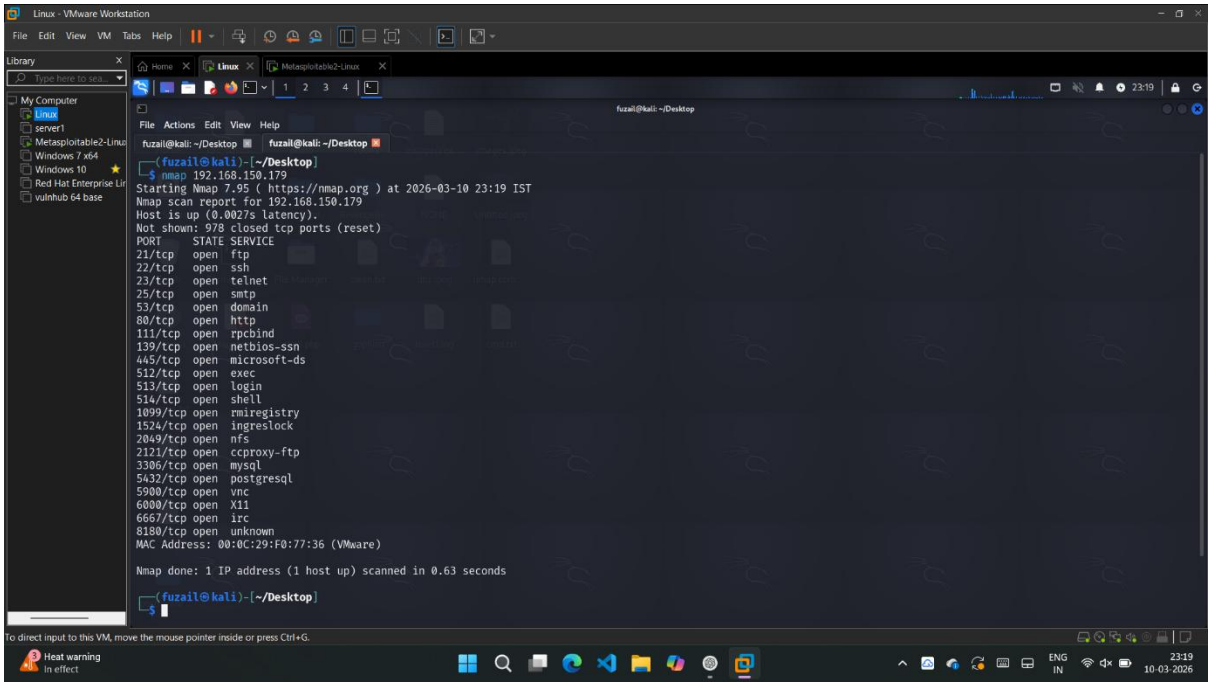


Figure 3: **SYN Stealth Scan**

Command: nmap -sS 192.168.150.179

Purpose: Performs a stealth scan to detect open ports. SYN stealth scan used for advanced port discovery.

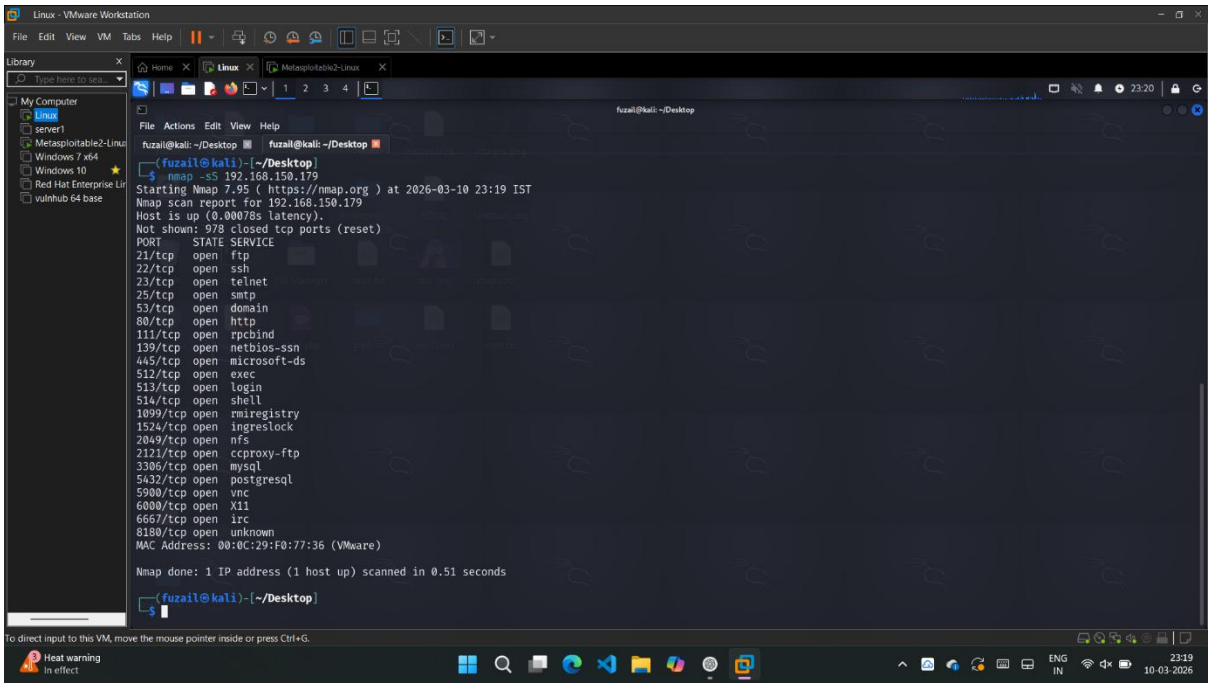


Figure 4: **Service Version Detection**

Command: nmap -sV 192.168.150.179

Purpose:
Detects the versions of services running on open ports. Service version detection using Nmap.

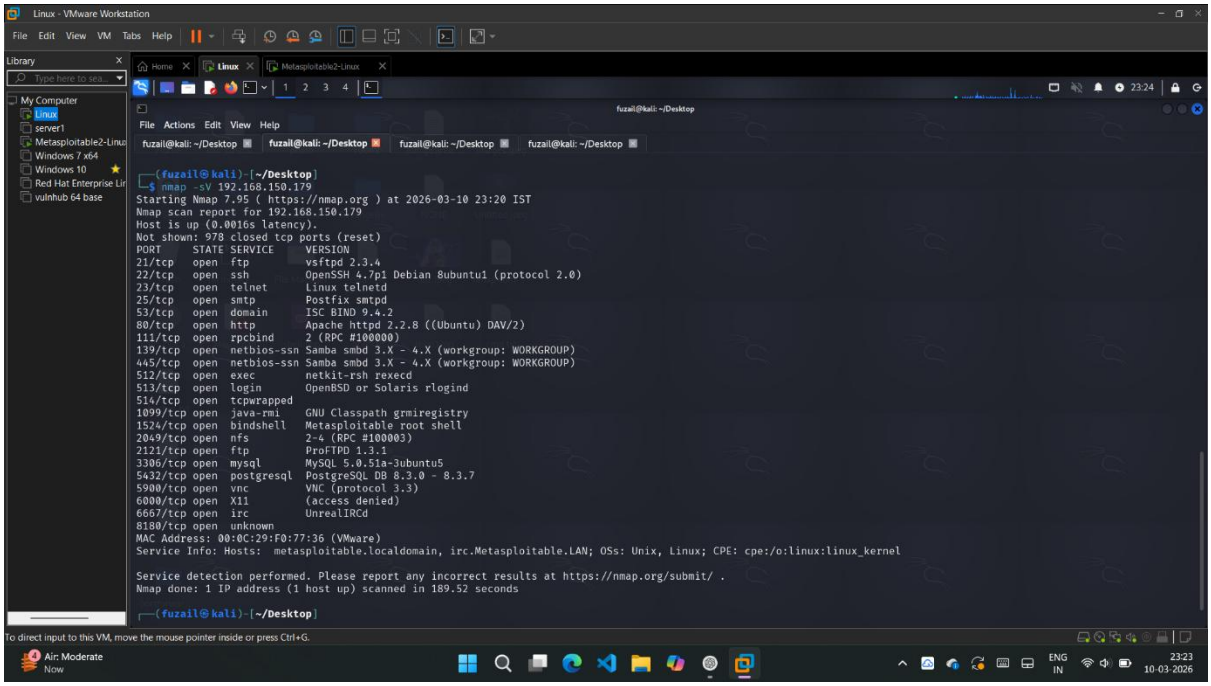


Figure 5: **Full Port Scan**

Command: sudo nmap -p- -T4 192.168.150.179

Purpose: Scans all 65,535 TCP ports.Full port scan identifying services running on non-standard ports

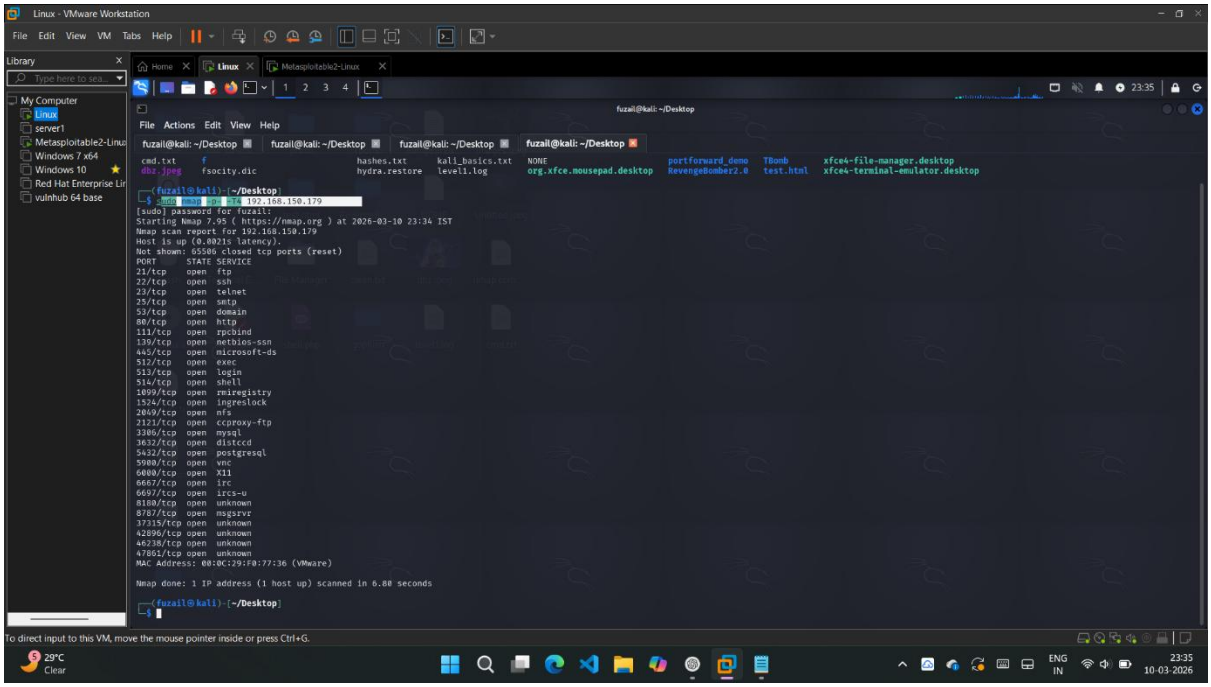


Figure 6: **Xmas Scan**

Command: sudo nmap -sX 192.168.150.179

Purpose: Uses special TCP flags to detect open or filtered ports. Xmas scan used for advanced port detection.

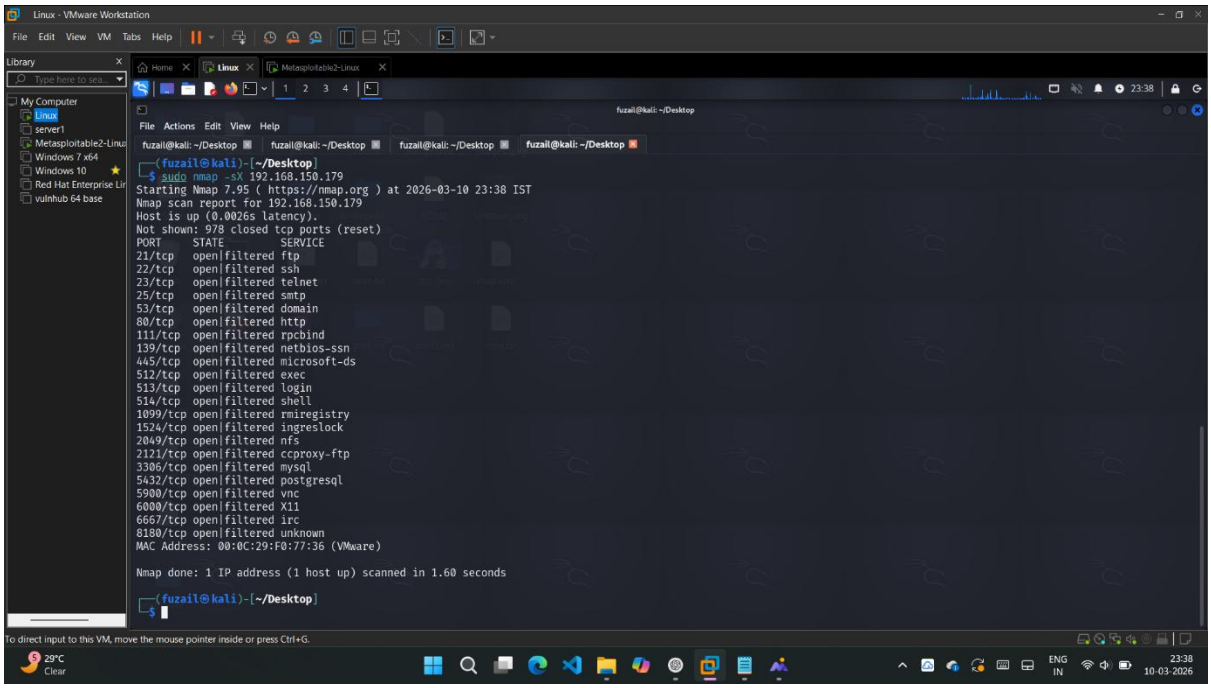


Figure 7:**Decoy Scan**

Command: sudo nmap -D RND:5 192.168.150.179

Purpose: Hides the real scanning IP address using decoy addresses. Decoy scan used to hide the origin of the scan.

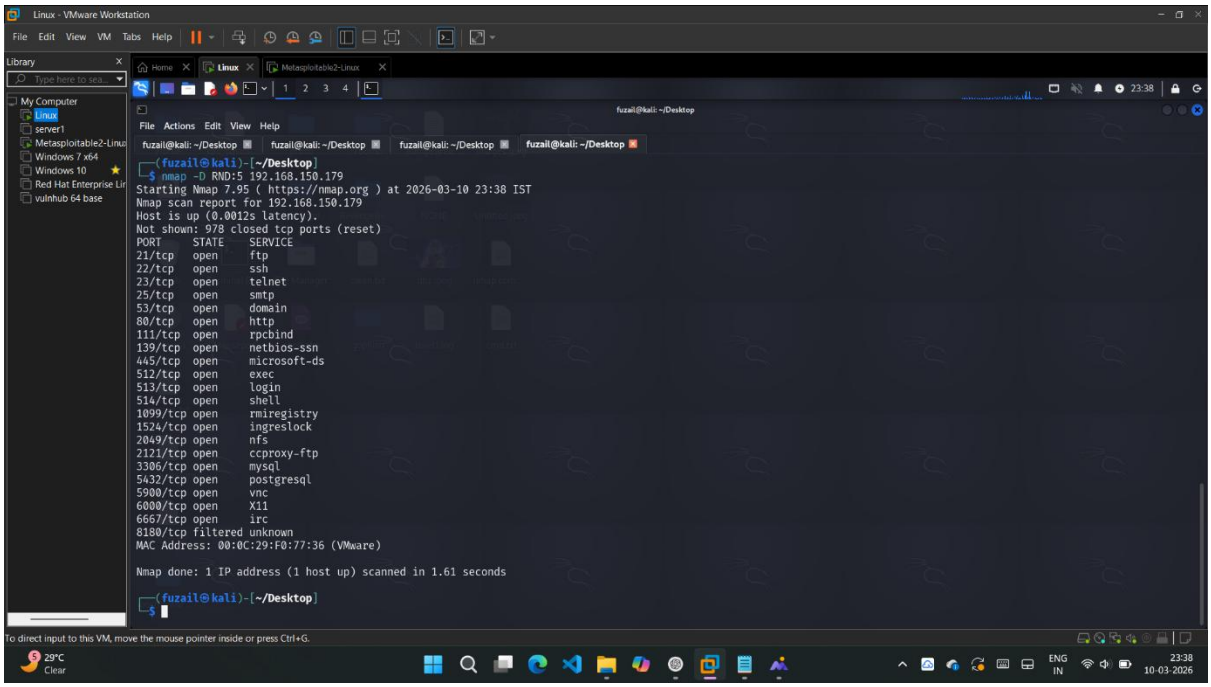


Figure 8: **Traceroute Scan**

Command: sudo nmap --traceroute 192.168.150.179

Purpose: Displays the network path to the target. Traceroute scan showing network path to the target system.

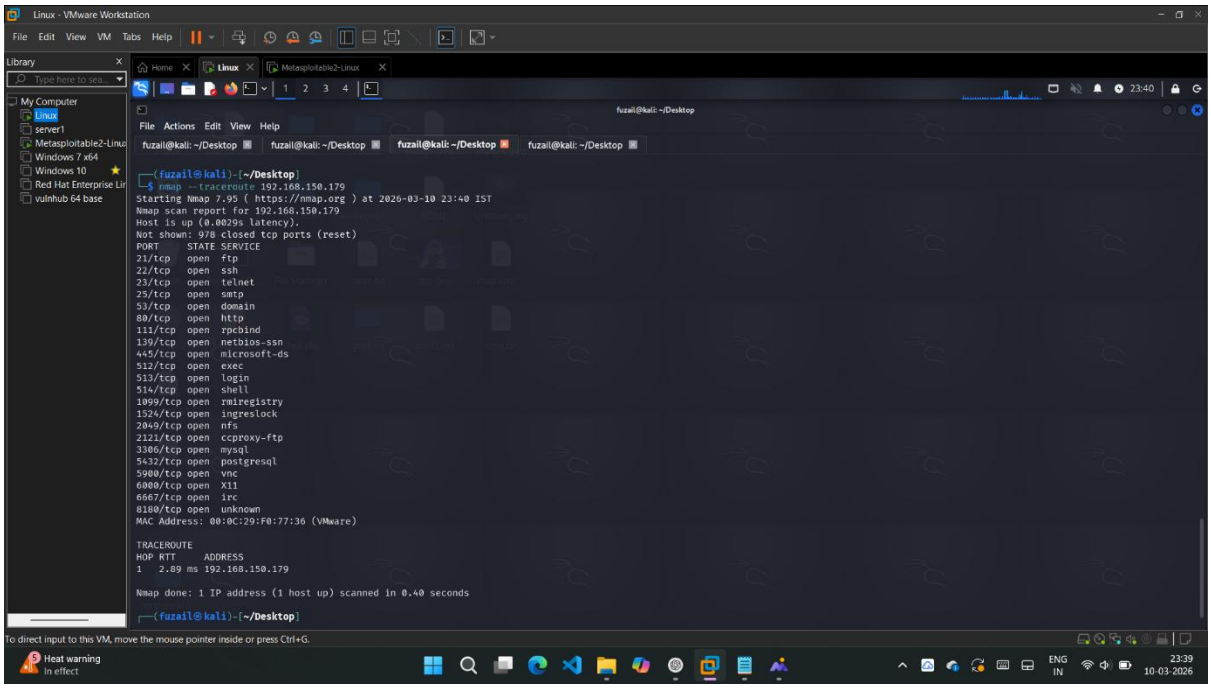


Figure 9: **Top Ports Scan**

Command: sudo nmap --top-ports 20 192.168.150.179

Purpose: Scans the 20 most commonly used ports. Scanning the most common network ports.

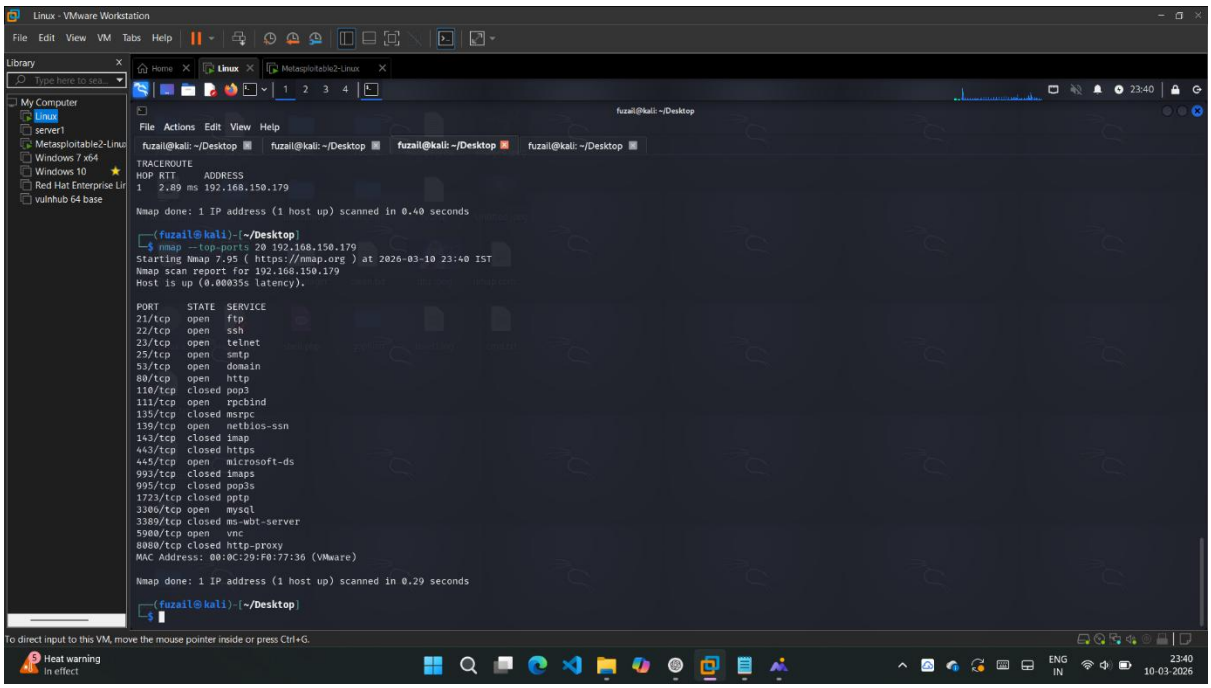


Figure 10: **Vulnerability Scan**

Command: sudo nmap --script vuln 192.168.150.179

Purpose: Runs vulnerability detection scripts. Running Nmap vulnerability scripts to detect security weaknesses.

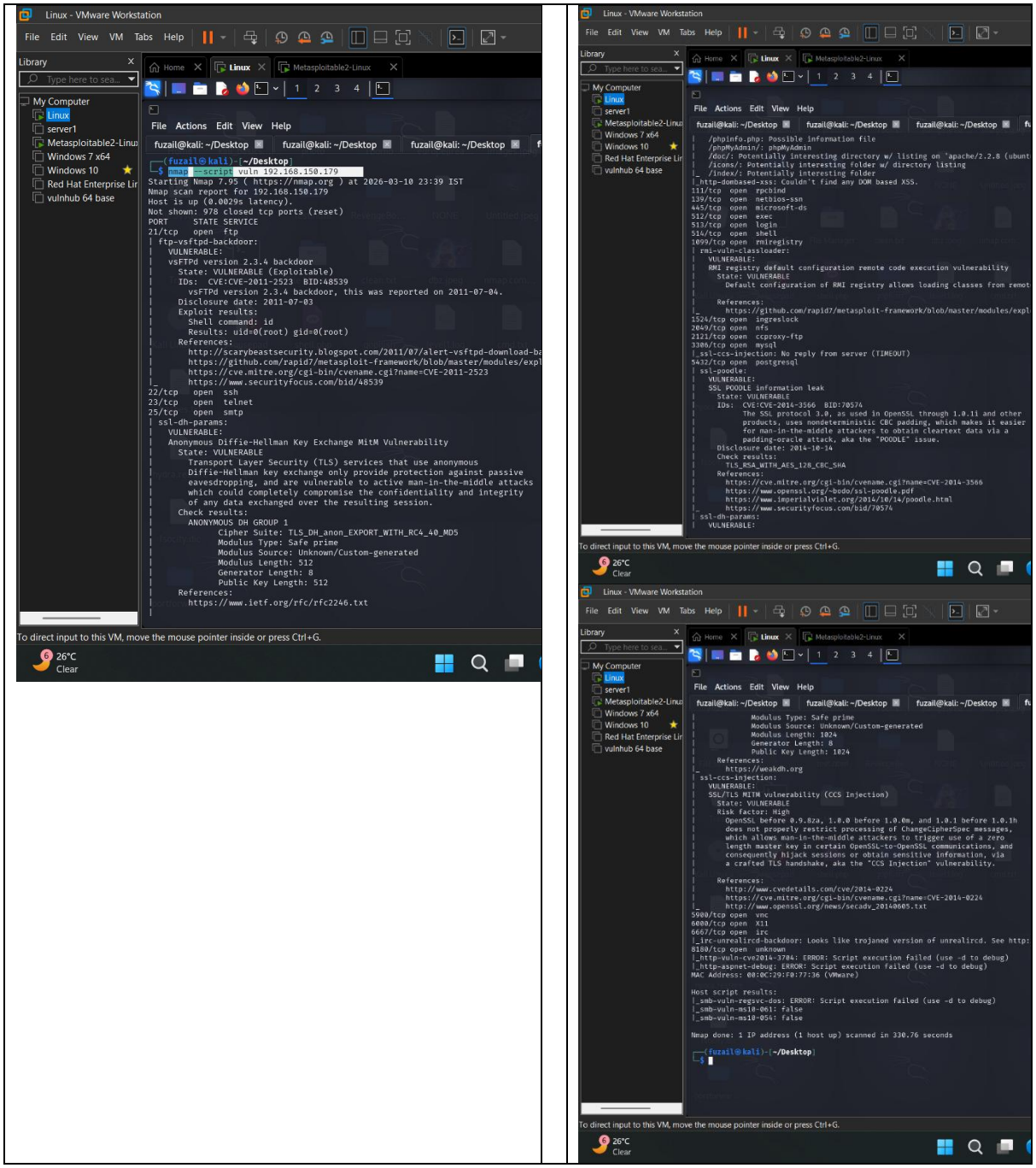


Figure 11: **Firewall Evasion Scan (Fragmented Packets)**

Command: sudo nmap -f 192.168.150.179

Explanation: -f Breaks packets into **small fragments**.

Why hackers use it: Some firewalls **fail to inspect fragmented packets**, allowing scans to bypass detection.

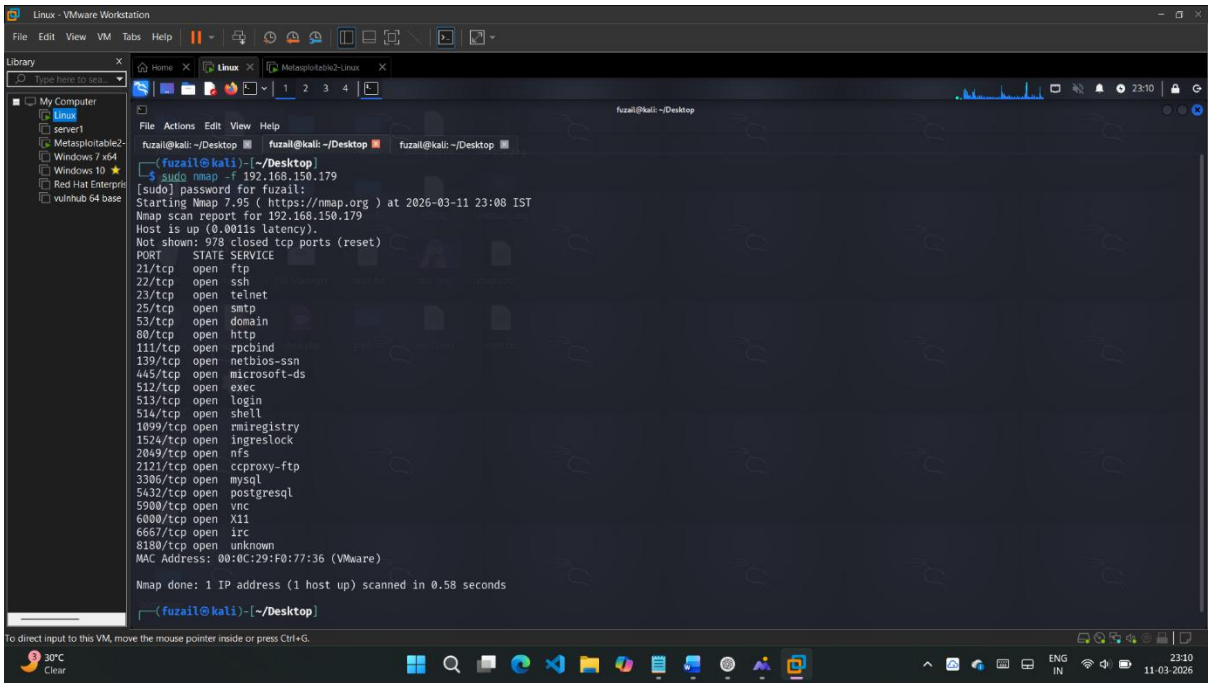


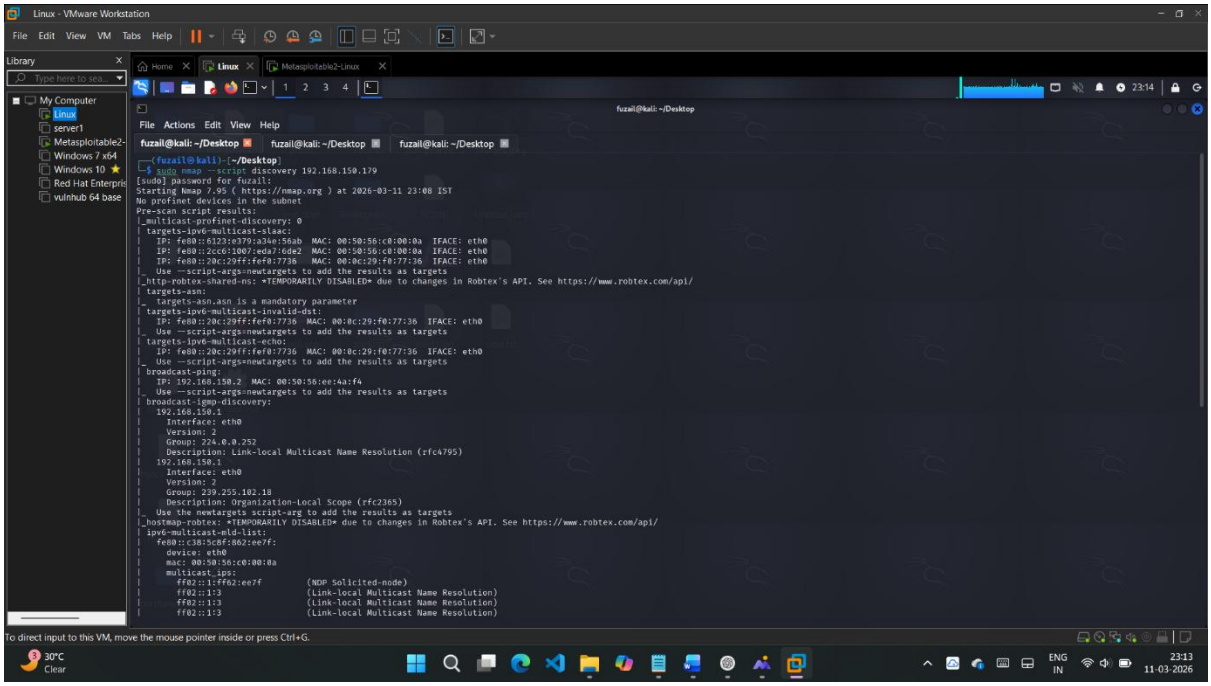
Figure 12: **NSE Script Discovery Scan**

Command: sudo nmap --script discovery 192.168.150.179

Explanation: Runs **Nmap Scripting Engine scripts** for host discovery and information gathering.

Why hackers use it: Collects extra information such as:

- open shares
- service details
- network information



7. Analysis

The Nmap scans revealed several open ports and running services on the target system. Services such as FTP, SSH, Telnet, HTTP, and database services were identified.any of these services are outdated or insecure, making the system vulnerable to attacks. For example, Telnet transmits credentials in plaintext, while older FTP services may allow anonymous login.

The presence of multiple open ports significantly increases the attack surface of the system.

8. Risk Assessment

Port	Service	Security Risk
21	FTP	Weak authentication
22	SSH	Brute force attacks
23	Telnet	Plaintext credentials
80	HTTP	Web vulnerabilities
3306	MySQL	Database exposure

9. Mitigation Strategies

To improve security, the following measures should be implemented:

- Disable unnecessary services
- Replace Telnet with SSH
- Use strong authentication mechanisms
- Apply regular security updates
- Configure firewall rules to restrict access

10. Conclusion

In this project, several advanced Nmap scanning techniques were performed to analyze the security posture of a target system. The scans successfully identified multiple open ports and running services on the Metasploitable machine.

The results demonstrate how network scanning tools can reveal potential vulnerabilities and help security professionals assess system security. Proper configuration, service management, and regular security updates are essential to reduce exposure to potential attack